STAT 6410
Homework 8

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1. **Chapter 4, Problem 3**

   Let ν be the counting measure on $[0,1]$ and $\int_{[0,1]} h d\nu < \infty$ for some nonnegative function h. Show that $B = \{x : h(x) > 0\}$ is countable.

   (**Hint:** Let $B_n = \{x : h(x) > \frac{1}{n}\}$. Show that B_n is a finite set for each $n \in \mathbb{N}$.)

   Ans: Clearly, $B_n = \{x \in [0,1] : h(x) > 0\} =\subseteq [0,1]$ for all $n \in \mathbb{N}$.
   Thus $\int_{B_n} h d\nu \leq \int_{[0,1]} h d\nu < \infty$ for all $n \in \mathbb{N}$.

   If $x \in B_n$, $\frac{1}{n} < h(x)$, so $\int \frac{1}{n} I_{B_n} < \int h I_{B_n} d\nu = \int_{B_n} h d\nu \leq \int_{[0,1]} h d\nu < \infty$.

   $\int \frac{1}{n} I_{B_n} d\nu = \frac{1}{n}\nu(B_n) < \infty \implies \nu(B_n) < \infty$ for all $n \in \mathbb{N}$.

   Since B_n is finite for every $n \in \mathbb{N}$ and $\cup_{n\geq 1} B_n = B$, B is countable.

2. **Chapter 4, Problem 4**

   Prove Proposition 4.1.2.

   Proposition 4.1.2: Let ν. μ, μ_1, μ_2, ... be σ-finite measure on a measurable space $(\Omega, \mathcal{F})$

   (i) If $\mu_1 << \mu_2$ and $\mu_2 << \mu_3$, then $\mu_1 << \mu_3$ and

   $$\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}\ a.e.(\mu_3)$$

   (ii) Suppose that μ_1 and μ_2 are dominated by μ_3. Then for any $\alpha, \beta \geq 0$, $\alpha\mu_1 + \beta\mu_2$ is dominated by μ_3 and

   $$\frac{d(\alpha\mu_1 + \beta\mu_2)}{d\mu_3} = \alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3} a.e.(\mu_3)$$

   (iii) If $\mu << \nu$ and $\frac{d\mu}{d\nu} > 0\ a.e.(\nu)$, then $\nu << \mu$ and

   $$\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1}\ a.e.(\mu).$$

(iv) Let $\{\mu_n\}_{n\geq 1}$ be a sequence of measure and $\{\alpha\}_{n\geq 1}$ be a sequence of positive real numbers, i.e., $\alpha > 0$ for all $n \geq 1$. Define $\mu = \sum_{n=1}^{\infty} \alpha_n \mu_n$

(a) Then $\mu << \nu$ iff $\mu_n << \nu$ for each $n \geq 1$ and in this case,

$$\frac{d\mu}{d\nu} = \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu} \; a.e.(\nu).$$

(b) $\mu \perp \nu$ iff $\mu_n \perp \nu$ for all $n \geq 1$

Ans: Lemma: For any σ-finite measures μ, ν on $(\Omega, \mathcal{F})$ with $\mu << \nu$, for any measurable function h:

$$\int h d\mu = \int h \frac{d\mu}{d\nu} d\nu$$

Proof. 1: Indicator Functions Let $h = \mathbf{I}_A$ for some $A \in \mathcal{F}$.

$$\int h d\mu = \int \mathbf{I}_A d\mu = \int_A d\mu$$
$$= \int \frac{d\mu}{d\nu} d\nu \qquad \text{(by RN Theorem, where } \tfrac{d\mu}{d\nu} \text{ is the RN der.)}$$
$$= \int \mathbf{I}_A \frac{d\mu}{d\nu} d\mu = \int h \frac{d\mu}{d\nu} d\nu$$

2: Simple Functions Let $\{A_1, A_2, ..., A_k\} \subset \mathcal{F}$ and $c_1, c_2, ..., c_k \in \mathbb{R}$. Define $h = \sum_{i=1}^{k} c_i \mathbf{I}_{A_i}$

$$\int h d\mu = \int \sum_{i=1}^{k} c_i \mathbf{I}_{A_i} d\mu = \sum_{i=1}^{k} c_i \int \mathbf{I}_{A_i} d\mu$$
$$= \sum_{i=1}^{k} c_i \int \mathbf{I}_{A_i} \frac{d\mu}{d\nu} d\nu \qquad (1)$$
$$= \int \sum_{i=1}^{k} c_i \mathbf{I}_{A_i} \frac{d\mu}{d\nu} d\nu = \int h \frac{d\mu}{d\nu} d\nu$$

3: Nonnegative Measurable Function, h, on $(\Omega, \mathcal{F})$: Define $\{h_n\}_{\{n\geq 1\}}$ is a sequence of nonnegative simple functions such that $h_n \uparrow h$.

$$\int h d\mu = \lim_{n\to\infty} \int h_n d\mu \qquad \text{(def of int. of nonneg. mbl. fn)}$$
$$= \lim_{n\to\infty} \int h_n \frac{d\mu}{d\nu} d\nu \qquad (2)$$
$$= \int h \frac{d\mu}{d\nu} d\nu \qquad \text{(MCT)}$$

4: Measurable Function on $(\Omega, \mathcal{F})$, h h can be written as a linear combination of 2 nonnegative measurable functions, h^+ and h^-. $h = h^+ - h^-$.
From (3), we know that
$\int h d\mu = \int h^+ d\mu - \int h^- d\mu = \int h^+ \frac{d\mu}{d\nu} d\nu - \int h^- \frac{d\mu}{d\nu} d\nu = \int h^+ - h^- \frac{d\mu}{d\nu} d\nu = \int h \frac{d\mu}{d\nu} d\nu$
□

(i)

Proof. Suppose, without loss of generality, that $\mu_3(A) = 0$ for some $A \in \mathcal{F}$. Thus, $\mu_2(A) = 0$, since $\mu_2 << \mu_3$. Finally, $\mu_1(A) = 0$, since $\mu_1 << \mu_2$. Therefore, $\mu_1 << \mu_3$.

$$\mu_1(A) = \int I_A \mu_1 = \int I_A \frac{d\mu_1}{d\mu_2} d\mu_2 \qquad \text{(R-N Theorem)}$$

From R-N theorem, we know that $I_A \frac{d\mu_1}{d\mu_2}$ is a mble nonneg fn

$$\mu_1(A) = \int I_A \frac{d\mu_1}{d\mu_2} d\mu_2 = \int I_A \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3} d\mu_3 \qquad \text{(from the Lemma)}$$

$\frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3}$ is the R-N derivative of μ_1 $w.r.t.$ mu_3

By R-N Theorem, $\mu_1(A) = \int I_A \frac{d\mu_1}{d\mu_3} d\mu_3$

From the definition of an R-N derivative, we know: $\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3} a.e.(\mu_3)$

□

(ii)

Proof. Because $\alpha, \beta \geq 0$, $\alpha\mu_1 + \beta\mu_2$ is σ-finite.
For all $A \in \mathcal{F}$

$$\begin{aligned} &\alpha\mu_1(A) + \beta\mu_2(A) = \alpha \int I_A d\mu_1 + \beta \int I_A d\mu_2 \\ &= \alpha \int I_A \frac{d\mu_1}{d\mu_3} d\mu_3 + \beta \int I_A \frac{d\mu_2}{d\mu_3} d\mu_3 \qquad \text{(R-N Theorem)} \\ &= \int_A \alpha \frac{d\mu_1}{d\mu_3} + \beta \frac{d\mu_2}{d\mu_3} d\mu_3 \end{aligned}$$

Thus, $\alpha \frac{\mu_1}{\mu_3} + \beta \frac{\mu_2}{\mu_3}$ is the R-N derivative for $\alpha\mu_1 + \beta\mu_2$ w.rt. μ_3

Since $\mu_1 << \mu_3$ and $\mu_2 << \mu_3$, $\mu_3(A) = 0 \implies \mu_2(A) = 0$, $\mu_1(A) = 0$. Thus, $\alpha\mu_1(A) + \beta\mu_2(A) = 0$. Thus, $\alpha\mu_1 + \beta\mu_2 << \mu_3$.

Thus, $\alpha\mu_1(A) + \beta\mu_2(A) = \int_A \frac{\alpha\mu_1+\beta\mu_2}{d\mu_3} d\mu_3$ for all $A \in \mathcal{F}$

Thus, $\frac{d(\alpha\mu_1+\beta\mu_2)}{d\mu_3}$ is the R-N derivative of $\alpha\mu_1 + \beta\mu_2$ w.r.t. μ_3.

By the definition of an R-N derivative, $\frac{d(\alpha\mu_1+\beta\mu_2)}{d\mu_3} = \alpha\frac{\mu_1}{\mu_3} + \beta\frac{\mu_2}{\mu_3} a.e.(\mu_3)$. □

(iii)

Proof. Suppose for some $A \in \mathcal{F}$ that $\mu(A) = 0$.
Thus, $0 = \int I_A \frac{d\mu}{d\nu} d\nu$, since $\mu << \nu$. From R-N theorem, we know that $\frac{d\mu}{d\nu}$ is measurable, and from the problem, we know that $\frac{d\mu}{d\nu} > 0$.

Thus, $0 = \int I_A \frac{d\mu}{d\nu} d\nu = \int I_A d\nu = \nu(A)$.

Thus, $\nu << \mu$.

For any $A \in \mathcal{F}$, define, $h \equiv I_A \left(\frac{d\mu}{d\nu}\right)^{-1}$. We know h is measurable and nonnegative from the R-N theorem ($\mu << \nu$) and because $\frac{d\nu}{d\nu} > 0 a.e.(\nu)$.

$$\nu(A) = \int I_A d\nu = \int I_A \frac{d\nu}{d\mu}\frac{d\mu}{d\nu} d\nu = \int I_A \left(\frac{d\mu}{d\nu}\right)^{-1} \frac{d\mu}{d\nu} d\nu$$
$$= \int I_A \left(\frac{d\mu}{d\nu}\right)^{-1} d\mu = \int h d\mu = \int I_A \left(\frac{d\mu}{d\nu}\right)^{-1} d\mu$$

$$\nu(A) = \int I_A \nu = \int I_A \frac{d\nu}{d\mu} d\mu$$

Since $\frac{d\nu}{d\mu}$ and $\left(\frac{d\mu}{d\nu}\right)^{-1}$ are both R-N derivatives of ν with respect to μ, $\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1} a.e.(\mu)$ □

(iv)

μ is clearly a valid measure on $(\Omega, \mathcal{F})$.

(a)

Proof. Suppose $\mu << \nu$. Suppose, that for some arbitrary $A \in \mathcal{F}$, $\nu(A) = 0$. Thus, $\mu(A) = 0 = \mu(\sum_{n=1}^{\infty} \alpha_n \mu_n(A)$. Since, $\mu_n(A) \geq 0$ and $\alpha_n > 0$ for all $n \in \mathbb{N}$, $\mu_n(A) = 0$ for all $n \geq 1$. Thus, $\mu_n << \nu$ for all $n \geq 1$.

Suppose $\mu_n << \nu$ for each $n \geq 1$. Suppose, that for some arbitrary $A \in \mathcal{F}$, $\nu(A) = 0$. Thus, for each $n \geq 1$, $\mu_n(A) = 0$. $\mu(A) = \sum_{n=1}^{\infty} \alpha_n \mu_n(A) = \sum_{n=1}^{\infty} \alpha_n \times 0 = 0$. Thus, $\mu << \nu$.

$$\mu(A) = \int_A \frac{d\mu}{d\nu} d\nu$$

$$\mu(A) = \sum_{n=1}^{\infty} \alpha\mu_n(A) = \sum_{n=1}^{\infty} \int_A \alpha_n \frac{d\mu_n}{d\nu} d\nu$$

From, R-N Theorem, we know that $\alpha_n \frac{d\mu_n}{d\nu}$ are nonnegative and measurable, so we can invoke MCT:

$$\mu(A) = \int_A \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu} d\nu$$

By the definition of an R-N derivative, $\frac{d\mu}{d\nu} = \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu} a.e.(\nu)$

Thus, $\mu << \nu \iff \mu_n << \nu$ for all $n \geq 1$ □

(b)

Proof. Suppose that $\mu \perp \nu$. Thus, there exists some set $B \in \mathcal{F}$ such that $\mu(B) = 0$ and $\nu(B^c) = 0$. As in (a), $\mu(B) = 0 \implies \mu_n(B) = 0$ for all $n \geq 1$.

Thus, $\mu_n \perp \nu$ for all $n \geq 1$.

Suppose that $\mu_n \perp \nu$ for all $n \geq 1$. Thus, there exists some collection of set $B_n \in \mathcal{F}$ such that $\mu_n(B_n) = 0$ and $\nu(B_n^c) = 0$ for all $n \geq 1$.

$\cap_{n\geq 1} B_n \in \mathcal{F}$. For each $n \geq 1$, $\cap_{n\geq 1} B_n \subseteq B_n$. Thus, $\mu_n(B_n) = 0 \implies \mu_n(\cap_{n\geq 1} B_n) = 0$. $\mu(\cap_{n\geq 1} B_n) = \sum_{n=1}^{\infty} \mu_n(\cap_{n\geq 1} B_n) = 0$

$(\cap_{n\geq 1} B_n)^c = \cup_{n\geq 1} B_n^c$

$\nu(\cup_{n\geq 1} B_n^c) \leq \sum_{n=1}^{\infty} \nu(B_n^c) = \sum_{n=1}^{\infty} 0 = 0$ (Countable subadditivity)

$\nu \perp \mu_n$ for all $n \geq 1$

Thus, $\nu \perp \mu \iff \nu \perp \mu_n$ for all $n \geq 1$. □

3. **Chapter 4, Problem 5** Find the Lebesgue decomposition of μ w.r.t ν and the Radon-Nikodym derivative $\frac{d\mu_a}{d\nu}$ in the following cases where μ_a is the absolutely continous component of μ w.r.t. ν.

   (a) $\mu = N(0,1)$, $\nu = Exponential(1)$

   (b) $\mu = Exponential(1)$, $\nu = N(0,1)$

   (c) $\mu = \mu_1 + \mu_2$, where $\mu_1 = N(0,1)$, $\mu_2 = Poisson(1)$ and $\nu = Cauchy(0,1)$.

   (d) $\mu = \mu_1 + \mu_2$, $\nu = Geometric(p)$, $0 < p < 1$,where $\mu_1 = N(0,1)$ and $\mu_2 =$ Poisson(1).

   (e) $\mu = \mu_1 + \mu_2$, $\nu = \nu_1 + \nu_2$ where $\mu_1 = N(0,1)$, $\mu_2 =$ Poisson(1), $\nu_1 =$ Cauchy(0, 1) and $\nu_2 =$ Geometric(p), $0 < p < 1$.

(f) μ = Binomial (10, 1/2), ν = Poisson (1). The measures referred to above are defined in Tables 4.6.1 and 4.6.2, given at the end of this section.

Ans: (a)

Define m to be the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, and $m_1(A) \equiv m(A \cap [0, \infty))$ for all $A \in \mathcal{B}(\mathbb{R})$, i.e. the Lebesgue measure restricted to positive numbers.

Define $\mu_a(A) \equiv \mu(A \cap [0, \infty))$ for all $A \in \mathcal{B}(\mathbb{R})$. In a similar manner to 4.1.1, we can show that $\mu_a << m_1$ and $m_1 << \nu$. Thus, by Proposition 4.1.2 (i), $\mu_a << \nu$.

Define $\mu_s \equiv \mu - \mu_a$. Thus, $\mu_s(A) = \mu(A \cap (-\infty, 0)$ for all $A \in \mathcal{B}(\mathbb{R})$. Define $B \equiv (-\infty, 0)$. $\nu(B) = 0$ and $\mu_s(B^c) = \mu_s([0, \infty)) = 0$. Thus, $\nu \perp \mu_s$. Thus, $\mu = \mu_a + \mu_s$ is the Lebesgue decomposition of μ w.r.t. ν.

$f_{\mu_a} = [N(0,1) \text{ pdf}]\ I_{x \in [0,\infty)}$
$f_\nu = Exp(1)$ pdf

$\frac{d\mu_a}{d\nu} = \frac{d\mu_a}{dm_1}\left(\frac{d\nu}{dm_1}\right)^{-1} = \frac{f_{\mu_a}}{f_\nu} = \frac{1}{\sqrt{2\pi}} e^{x - \frac{x^2}{2}} I_{x \in [0,\infty)} a.e.(\nu)$

(b) Define m and m_1 as in (a).

$\mu << m_1 << m$ and from, Example 4.1.1, $m << \nu$. By Proposition 4.1.2 (i), $\mu << \nu$. So, $\mu_a = \mu$. Define $\mu_s \equiv \mu - \mu_a = 0$. μ_s is the 0 measure, so clearly, $\mu_s \perp \nu$. Therefore the Lebesgue decomposition of μ w.r.t. ν is $\mu = \mu_a + \mu_s = \mu$.

$f_{\mu_a} = Exp(1)$ pdf

$f_\nu = N(0,1)$ pdf. $\frac{d\mu_a}{d\nu} = \frac{d\mu_a}{dm}\left(\frac{d\nu}{dm}\right)^{-1} = \frac{f_{\mu_a}}{f_\nu} = \sqrt{2\pi} e^{x + \frac{x^2}{2}} I_{x \in [0,\infty)} a.e.(\nu)$

(c) From before, we know that $\mu_1 << m$, and since the support of $Cauchy(0,1)$

Since μ_2 is discrete and ν is continuous $\mu_2 \perp \nu$. So, the Lebesgue decomposition of μ w.r.t. ν is $\mu = \mu_a + \mu_s$.

$f_\nu = Cauchy(0,1)$ pdf
$f_{\mu_a} = N(0,1)$ pdf.

$\frac{\mu_a}{\nu} = \frac{d\mu_a}{dm}\left(\frac{d\nu}{dm}\right)^{-1} = \frac{f_{\mu_a}}{f_\nu} = \sqrt{\frac{\pi}{2}}(1 + x^2) e^{x^2/2} a.e.(\nu)$

(d) Define λ as the counting measure $\mu_1 \perp \nu$ because $\mu_1(\mathbb{N}) = 0$ and $\nu(\mathbb{N}^c) = 0$ $\mu_2 << \nu$ because $\mu_2 << \lambda$, $\lambda << \nu$, and this implies $\mu_2 << \nu$

Thus, $\mu_s = \mu_1$ and $\mu_a = \mu_2$

$\frac{d\mu_a}{d\nu}(\omega) = \frac{d\mu_a}{d\lambda}(\frac{d\nu}{d\lambda})^{-1} = \frac{\sum_{\omega \in \mathbb{N}} I_{\omega = k} \frac{1}{k!e}}{\sum_{\omega \in \mathbb{N}} I_{\omega = k} p(1-p)^k} a.e.(\nu)$

(e) $\mu_1 << \nu$, because $\mu_1 << m$ and $m << \nu_1$ which implies $m << \nu$ which implies $\mu_1 << \nu$ Similarly, $\mu_2 << \nu$, because $\mu_2 << \lambda$ and $\lambda << \nu_2$ which implies $\lambda << \nu$ which implies $\mu_1 << \nu$

By Proposition 4.1.2 (d), $\mu_1 + \mu_2 = \mu << \nu$

Trivially, $0 \perp \nu$ (where 0 is the zero measure)

So, $\mu_s = 0$ and $\mu_a = \mu$

Additionally, it is important to note that $\mu_1 \perp \nu_2$ and $\mu_2 \perp \nu_2$

From Proposition 4.1.2, for all $A \in \mathcal{F}$,

$$\begin{aligned}
&\mu(A) = \mu_1(A) + \mu_2(A) \\
&= \int_A \frac{d\mu_1}{d\nu_1} d\nu_1 + \int_A \frac{d\mu_2}{d\nu_2} d\nu_2 \\
&= \int_{A\cap\mathbb{N}^c} \frac{d\mu_1}{d\nu_1} d\nu_1 + 0 + \int_{A\cap\mathbb{N}} \frac{d\mu_2}{d\nu_2} d\nu_2 + 0 \\
&= \int_{A\cap\mathbb{N}^c} \frac{d\mu_1}{d\nu_1} d(\nu_1+\nu_2) + \int_{A\cap\mathbb{N}} \frac{d\mu_2}{d\nu_2} d(\nu_1+\nu_2)
\end{aligned}$$

Thus, $\frac{d\mu_a}{d\nu}(\omega) = \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c}(\omega) + \frac{d\mu_2}{d\nu_2} I_{\mathbb{N}}(\omega) = \sqrt{\frac{\pi}{2}}(1+x^2)e^{-\omega^2/2} I_{\omega\in\mathbb{N}^c} + \sum_{\omega\in\mathbb{N}} \frac{1}{e\omega! p(1-p)^\omega} I_{\omega\in\mathbb{N}}$

(f) The support of μ is $\{0, 1, 2, ..., 10\}$ and the support of ν is $\mathbb{N} \cap \{0\}$. Clearly, $\mu << \nu$

Thus, $\mu_a = \mu$ and $\mu_s = 0$, where $\nu = \mu_a + \mu_s$ For some $A \in \mathcal{F}$,
$\mu(A) = \sum_{\omega\in A\cap\{0,1,...,10\}} \binom{10}{\omega}(1/2)^{10} = \int_A \frac{d\mu}{d\nu} d\nu = \int_{A\cap\{0,1,...,10\}} \binom{10}{\omega}(1/2)^{10} \frac{1}{e\omega!} d\nu(\omega)$

Thus, $\frac{d\mu}{d\nu} = \binom{10}{\omega}(1/2)^{10} \frac{1}{e\omega!} I_{\{0,1,...,10\}}(\omega) a.e.(\nu)$

4. **Additional Problem**

Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ-finite measures on it. Define the set function $\mu_3 \equiv \mu_1 + \mu_2$ (i.e. for all $A \in \mathcal{F}$, $\mu_3(A) = \mu_1(A) + \mu_2(A)$. Then prove the following:

(a) μ_3 is a measure

(b) if $\mu_1 << \nu$ and $\mu_2 << \nu$, then $\mu_3 << \nu$

(c) Express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

(d) If $\mu << \nu$ and $\mu \perp \nu$ holds, show that μ is the zero measure.

(e) If $\mu_1 = Lognormal(5, 6), \mu_2 = Geometric(0.3)$, and $\nu = Poisson(8)$ on $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then find the Lebesgue decomposition of μ_3 w.r.t ν, and find the Radon Nikodym derivative $\frac{d\mu_{3a}}{d\nu}$, where μ_{3a} is the absolutely continuous component of μ_3 in the decomposition.

(a)

Proof. (i) Let $A \in \mathcal{F}$ be given. $\mu_1(A) \in [0, \infty]$ and $\mu_2(A) \in [0, \infty]$, so $\mu_3(A) = \mu_1(A) + \mu_2(A) \in [0, \infty]$

(ii) $\mu_1(\emptyset) = 0$ and $\mu_2(\emptyset) = 0$, so $\mu_3(\emptyset) = \mu_1(\emptyset) + \mu_2(\emptyset) = 0$

(iii) Let $A_1, A_2, ... \in \mathcal{F}$ be disjoint and $\cup_{n\geq 1} A_n \in \mathcal{F}$.
$\mu_3(\cup_{n\geq 1} A_n) = \mu_1(\cup_{n\geq 1} A_n) + \mu_2(\cup_{n\geq 1} A_n) = \sum_{n=1}^{\infty} \mu_1(A_n) + \sum_{n=1}^{\infty} \mu_2(A_n) = \sum_{n=1}^{\infty} (\mu_1(A_n) + \mu_2(A_n)) = \sum_{n=1}^{\infty} \mu_3(A)$

Thus, μ_3 is a measure on $(\mathcal{F}, \Omega)$ □

(b)

Proof. Let $A \in \mathcal{F}$ be an arbitrary set such that $\nu(A) = 0$. Since $\mu_1 << \nu$, $\mu_1(A) = 0$. Similarly, $\mu_2(A) = 0$. Thus, $\mu_3(A) = \mu_1(A) + \mu_2(A) = 0 + 0 = 0$.

Thus, $\nu(A) = 0 \implies \mu_3(A) = 0$, and thus $\mu_3 << \nu$ □

(c)

Proof. The R-N derivative of μ_1 w.r.t. ν, denoted by $\frac{d\mu_1}{d\nu}$ is a nonnegative function such that $\mu_1(A) = \int_A \frac{d\mu_1}{d\nu} d\nu \; \forall \; A \in \mathcal{F}$.
The R-N derivative of μ_2 w.r.t. ν, denoted by $\frac{d\mu_2}{d\nu}$ is a nonnegative function such that $\mu_2(A) = \int_A \frac{d\mu_2}{d\nu} d\nu \; \forall \; A \in \mathcal{F}$.
The R-N derivative of μ_3 w.r.t. ν, denoted by $\frac{d\mu_3}{d\nu}$ is a nonnegative function such that $\mu_3(A) = \int_A \frac{d\mu_3}{d\nu} d\nu \; \forall \; A \in \mathcal{F}$.

For all $A \in \mathcal{F}$, $\mu_3(A) = \mu_1(A) + \mu_2(A) = \int_A \frac{d\mu_1}{d\nu} d\nu + \int_A \frac{d\mu_2}{d\nu} d\nu$
$= \int_A \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu} d\nu = \int_A \frac{d\mu_3}{d\nu} d\nu$

Thus, $\frac{d\mu_3}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$. □

(d)

Proof. Since $\mu \perp \nu$, there must exist some $B \in \mathcal{F}$ such that $\nu(B) = 0$ and $\mu(B^c) = 0$. Since $\mu << \nu$, this means that $\mu(B) = 0$.

$B, B^c \in \mathcal{F}$ and $B \cap B^c =$, so, by the def of a measure, $\mu(B) + \mu(B^c) = \mu(B \cup B^c) = \mu(\Omega) = 0$.

$A \subset \Omega$ for all $A \in \mathcal{F}$, so by monotonicity, $\mu(A) \leq \mu(\Omega) = 0$. Since $\mu(A) \in [0, \infty]$, $\mu(A) = 0$ for all $A \in \mathcal{F}$. Thus, μ is the zero measure. □

(e)

Measure	**Support**	$g(x)$	$f(x)$
μ_1	$(0,\infty)$	$g_{\mu_1}(x) = \frac{1}{x\sqrt{12\pi}} \exp\left(-\frac{(\log(x)-6)^2}{12}\right)$	$f_{\mu_1} = g_{\mu_1}(x) I_{x\in(0,\infty)}$
μ_2	$\mathbb{Z}_+$	$g_{\mu_2}(x) = 1 - (0.7)^{x+1}$	$f_{\mu_2}(x) = g_{\mu_2}(x) I_{x\in\mathbb{Z}_+}$
ν	$\mathbb{Z}_+$	$g_\nu(x) = \frac{8^x e^{-8}}{x!}$	$f_\nu(x) = g_\nu(x) I_{x\in\mathbb{Z}_+}$

$\lambda =$ the counting measure

If we define

$\mu_2 << \nu$ and $\mu_1 \perp \nu$, so $\mu_{3a} = \mu_2$. $\mu_2 << \lambda << \nu$, thus, by Proposition 4.1.2, $\frac{d\mu_{3a}}{d\nu} = \frac{d\mu_{3a}}{d\lambda}\frac{d\lambda}{d\nu} = \frac{d\mu_{3a}}{d\lambda}\frac{d\nu}{d\lambda})^{-1} = \frac{f_{\mu_2}}{f_\nu}$